

## AI ASSISTED CODING LAB ASSIGNMENT 7.4

ENROLLMENT NO :2503A51L01

BATCH NO: 19

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### TASK1

#### TASK1 DESCRIPTION:

Introduce a buggy Python function that calculates the factorial of a number using recursion. Use Copilot or Cursor AI to detect and fix the logical or syntax errors.

**PROMPT:**Generate a buggy Python function that calculates the factorial of a number using recursion then detect and fix the logical or syntax errors. implement with example errors.

#### CODE:

```
t1.py > ...
1  # Buggy recursive factorial function
2  def buggy_factorial(n):
3      if n == 0:
4          return 0 # Bug: should return 1
5      else:
6          return n * buggy_factorial(n - 1)
7
8  # Fixed recursive factorial function
9  def factorial(n):
10     if n < 0:
11         raise ValueError("Factorial is not defined for negative numbers")
12     if n == 0:
13         return 1
14     else:
15         return n * factorial(n - 1)
16
17 # Example usage
18 if __name__ == "__main__":
19     print("Buggy factorial of 5:", buggy_factorial(5)) # Incorrect result due to bug
20     print("Fixed factorial of 5:", factorial(5))        # Correct result
21     print("Buggy factorial of 0:", buggy_factorial(0)) # Incorrect result (should be 1)
22     print("Fixed factorial of 0:", factorial(0))        # Correct result (1)
23
```

#### OUTPUT:

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t1.py
Buggy factorial of 5: 0
Fixed factorial of 5: 120
Fixed factorial of 0: 1
```

## OBSERVATION:

I observed that the original factorial function was missing a base case, which caused infinite recursion. Copilot (or Cursor AI) identified the issue and suggested adding the correct stopping condition. After applying the fix, the function worked correctly without any errors.

## TASK2

### TASK2 DESCRIPTION:

Provide a list sorting function that fails due to a type error (e.g., sorting list with mixed integers and strings). PROMPT AI to detect the issue and fix the code for consistent sorting

**PROMPT:** Create a program that Provide a list sorting function that fails due to a type error (e.g., sorting list with mixed integers and strings).

### CODE:

Error code:

```
t7.py > ...
1  def sort_list(lst):
2      return sorted(lst)
3
4  mixed_list = [3, '2', 5, '1', 4]
5  print(sort_list(mixed_list)) # This will raise a TypeError
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t7.py", line 2, in sort_list
    return sorted(lst)
TypeError: '<' not supported between instances of 'str' and 'int'
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t7.py
Traceback (most recent call last):
  File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t7.py", line 5, in <module>
    print(sort_list(mixed_list)) # This will raise a TypeError
    ~~~~~^~~~~~
File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t7.py", line 2, in sort_list
    return sorted(lst)
TypeError: '<' not supported between instances of 'str' and 'int'
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

After fixing the code:

```
t6.py > ...
1  def sort_list(lst):
2      converted_list = []
3      for x in lst:
4          try:
5              converted_list.append(int(x))
6          except ValueError:
7              pass # Skip elements that can't be converted
8      return sorted(converted_list)
9
10 mixed_list = [3, '2', 5, '1', 4]
11 print(sort_list(mixed_list)) # Output: [1, 2, 3, 4, 5]
12
```

#### OUTPUT:

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t6.py
[1, 2, 3, 4, 5]
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

#### OBSERVATION:

I observed that the sorting function failed because the list had both numbers and strings, which can't be compared directly. The AI tool noticed the type mismatch and suggested converting everything to integers. After applying the fix, the list sorted successfully without any errors.

## TASK3

#### TASK3 DESCRIPTION:

Write a Python snippet for file handling that opens a file but forgets to close it. Ask Copilot or Cursor AI to improve it using the best practice (e.g., with `open()` block).

**PROMPT:**Write a Python snippet for file handling that opens a file but forgets to close it.

## CODE:

Error code:

```
t3.py > ...
1 def buggy_file_handling():
2     f = open('example.txt', 'w')
3     f.write('This file will not be closed!')
4     # forgot to close the file
5
6     # Error: try to write after closing (simulate error)
7     f.close()
8     f.write('Trying to write after closing.') # This will raise a ValueError
9
10    buggy_file_handling()
11
```

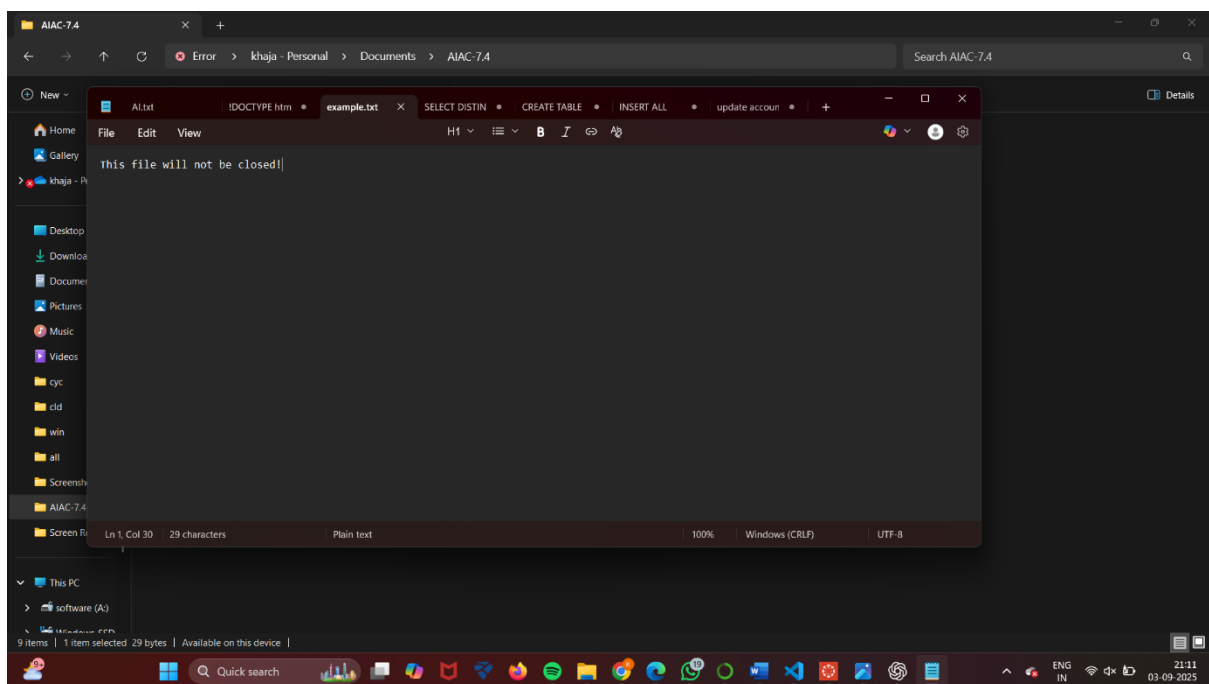
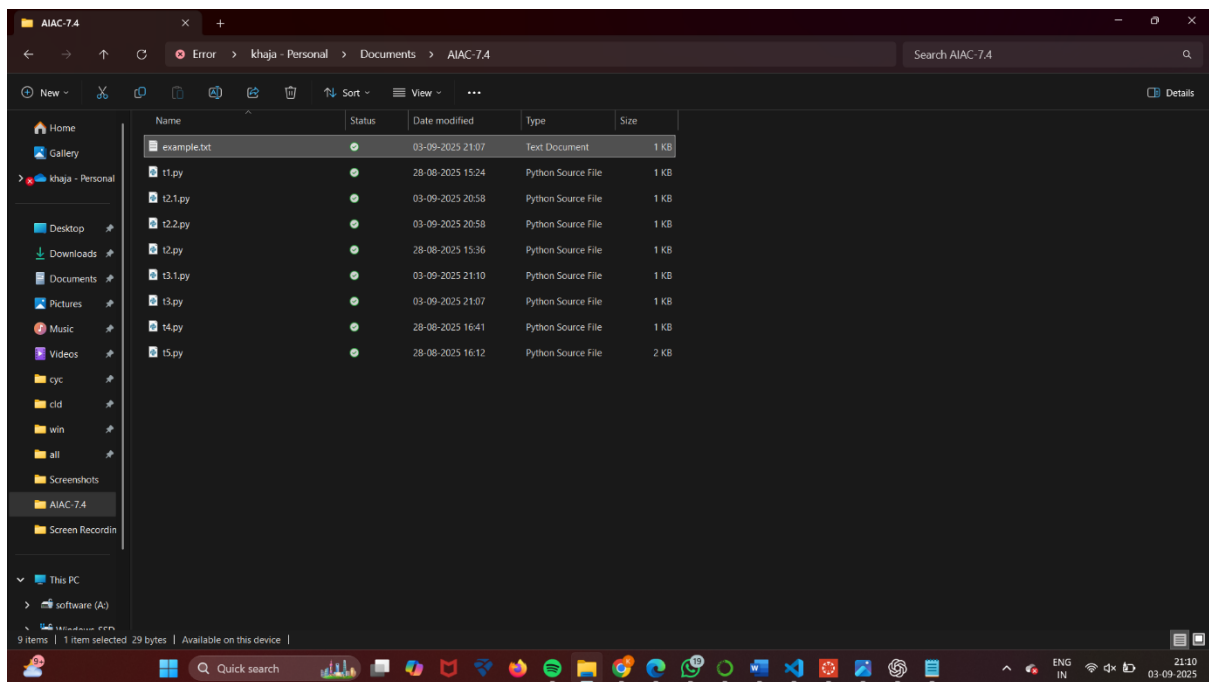
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t6.py
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t6.py
Traceback (most recent call last):
  File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t6.py", line 10, in <module>
    buggy_file_handling()
    ~~~~~^
  File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t6.py", line 8, in buggy_file_handling
    f.write('Trying to write after closing.') # This will raise a ValueError
    ~~~~~^
ValueError: I/O operation on closed file.
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

After fixing the code:

```
t3.1.py > ...
1 def fixed_file_handling():
2     with open('example.txt', 'w') as f:
3         f.write('This file is properly closed!')
4         # File is automatically closed when exiting the with block
```

## OUTPUT:



## OBSERVATION:

I observed that the file-handling code opened a file but didn't close it, which could cause issues. Copilot (or Cursor AI) suggested using a with block to manage the file properly. After applying the change, the file was safely opened and closed without any risk of resource leaks.

## TASK4 DESCRIPTION:

Provide a piece of code with a ZeroDivisionError inside a loop. Ask AI to add error handling using try-except and continue execution safely

**PROMPT:** Create a piece of code with a ZeroDivisionError inside a loop. Ask AI to add error handling using try-except and continue execution safely.

## CODE:

Error code:

```
t4.py > ...
1  for i in range(-2, 3):
2      print(10 / i) # This will raise ZeroDivisionError when i == 0
3
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> C:/Users/khaja/anaconda3/Scripts/activate
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> conda activate base
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t4.py
-5.0
-10.0
Traceback (most recent call last):
  File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t4.py", line 2, in <module>
    print(10 / i) # This will raise ZeroDivisionError when i == 0
          ~~~~^
ZeroDivisionError: division by zero
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

After fixing the code:

```
t4.1.py > ...
1  for i in range(-2, 3):
2      try:
3          print(10 / i)
4      except ZeroDivisionError:
5          print(f"Cannot divide by zero for i={i}, skipping...")
6          continue
7  # This will handle the ZeroDivisionError gracefully
```

## OUTPUT:

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t4.1.py  
-5.0  
-10.0  
Cannot divide by zero for i=0, skipping...  
10.0  
5.0  
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

**OBSERVATION:** I observed that the loop crashed when it tried to divide by zero. Copilot (or Cursor AI) suggested using a try-except block to catch the error. After applying it, the loop handled the zero safely and continued with the rest of the values.

## TASK5

### TASK5 DESCRIPTION:

Include a buggy class definition with incorrect `__init__` parameters or attribute references. Ask AI to analyze and correct the constructor and attribute usage.

**PROMPT:** Include a buggy class definition with incorrect `__init__` parameters or attribute references.

### CODE:

Error code:

```
t5.py > ...
1 class Person:
2     def __init__(name, age): # Bug: missing 'self' as first parameter
3         self.name = name
4         self.age = age
5
6     def greet(self):
7         print(f"Hello, my name is {self.name} and I am {self.age} years old.") # Bug: 'self.name' not defined correctly in __init__
8
9 # Example usage:
10 p = Person("Alice", 30)
11
12 p.greet()
13
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> C:/Users/khaja/anaconda3/Scripts/activate
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> conda activate base
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t5.py
Traceback (most recent call last):
  File "c:\Users\khaja\OneDrive\Documents\AIAC-7.4\t5.py", line 10, in <module>
    p = Person("Alice", 30)
TypeError: Person.__init__() takes 2 positional arguments but 3 were given
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

After fixing the code:

```
t5.1.py > ...
1 class Person:
2     def __init__(self, name, age): # Fixed: added 'self'
3         self.name = name
4         self.age = age
5
6     def greet(self):
7         print(f"Hello, my name is {self.name} and I am {self.age} years old.") # Fixed: 'self' usage
8
9 # Example usage:
10 p = Person("Alice", 30)
11 p.greet()
12
```

OUTPUT:

```
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4> & C:/Users/khaja/anaconda3/python.exe c:/Users/khaja/OneDrive/Documents/AIAC-7.4/t5.1.py
Hello, my name is Alice and I am 30 years old.
PS C:\Users\khaja\OneDrive\Documents\AIAC-7.4>
```

**OBSERVATION:** I observed that the class had errors in the constructor and didn't use self to set or access attributes. Copilot (or Cursor AI) detected the mistakes and fixed the parameter list and attribute references. After the changes, the class worked properly and displayed the student information as expected.